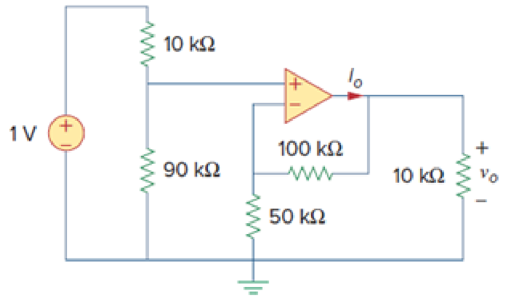
Problem

Find *vo* and *io* in the circuit of Fig. 5.52.

Figure. 5.52:

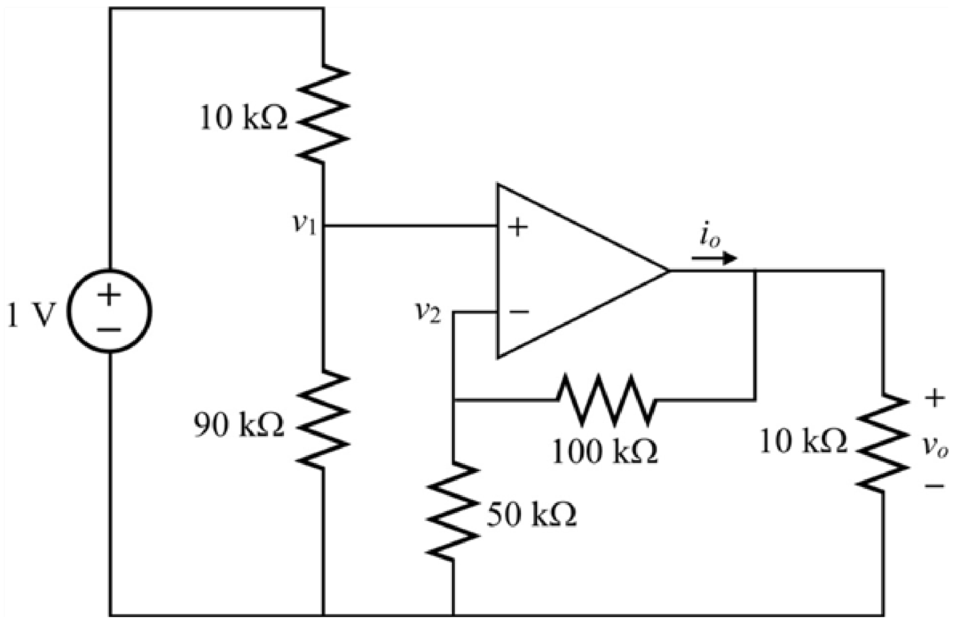


Step-by-step solution

Step 1 of 4

Refer to circuit diagram in Figure 5.52 in the textbook.

Draw the circuit diagram with node notations.



Step 2 of 4

Apply Kirchhoff's current law at node v_1.

$$\frac{v_1 - 1\text{ V}}{10\text{ k}\Omega} + \frac{v_1}{90\text{ k}\Omega} = 0$$
$$\left(\frac{1}{10\text{ k}\Omega} + \frac{1}{90\text{ k}\Omega}\right)v_1 = \frac{1\text{ V}}{10\text{ k}\Omega}$$
$$\left(\frac{9+1}{90\text{ k}\Omega}\right)v_1 = \frac{1\text{ V}}{10\text{ k}\Omega}$$
$$\begin{aligned} v_1 &= \left(\frac{1\text{ V}}{10\text{ k}\Omega}\right)\left(\frac{90\text{ k}\Omega}{10}\right) \\ &= 0.9\text{ V} \end{aligned}$$

It is known that, the voltages at inverting and non-inverting terminals of the op-amp are same for ideal op-amp.

$$\begin{aligned} v_1 &= v_2 \\ &= 0.9\text{ V} \end{aligned}$$

Step 3 of 4

Apply Kirchhoff's current law at node v_2.

$$\frac{v_2}{50\text{ k}\Omega} + \frac{v_2 - v_o}{100\text{ k}\Omega} = 0$$
$$\frac{v_o}{100\text{ k}\Omega} = \left(\frac{1}{50\text{ k}\Omega} + \frac{1}{100\text{ k}\Omega}\right)v_2$$
$$\frac{v_o}{100\text{ k}\Omega} = \left(\frac{2+1}{100\text{ k}\Omega}\right)v_2$$
$$v_o = 3v_2$$

Substitute 0.9 V for v_2.

$$\begin{aligned} v_o &= 3(0.9\text{ V}) \\ &= 2.7\text{ V} \end{aligned}$$

Therefore, the value of output voltage, v_o is $\boxed{2.7\text{ V}}$.

Step 4 of 4

Apply Kirchhoff's current law at node v_o.

$$i_o = \frac{v_o - v_2}{100\text{ k}\Omega} + \frac{v_o}{10\text{ k}\Omega}$$

Substitute 2.7 V for v_o and 0.9 V for v_2.

$$\begin{aligned} i_o &= \frac{2.7\text{ V} - 0.9\text{ V}}{100\text{ k}\Omega} + \frac{2.7\text{ V}}{10\text{ k}\Omega} \\ &= \frac{1.8\text{ V} + 27\text{ V}}{100\text{ k}\Omega} \\ &= \frac{28.8\text{ V}}{100\text{ k}\Omega} \\ &= 0.288\text{ mA} \end{aligned}$$

Therefore, the current, i_o is $\boxed{0.288\text{ mA}}$.